

MEDIATEK

MT8676 Android Camera Driver User Manual

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版本记录

版本	日期	作者	描述
1.0	2024-08-12	Jianmin Zhou	正式版
1.1	2024-10-15	Jianmin Zhou	在 1.3.1.9 配置 static_ctx 中增加 multicam 配置信息
1.2	2025-06-13	Jianmin Zhou	1.在 1.3.1.5 修改 dts 文件中增加非 JSON 化 sensor 的配置字段 2.在 1.3.1.6 增加新的 Meta 文件中增加 JSON 化 yuv sensor metadata 配置 3.在 1.3.1.9 配置 static_ctx 和 1.3.1.13 配置 get_sensor_usage 函数中增加 multicam_async 模式配置

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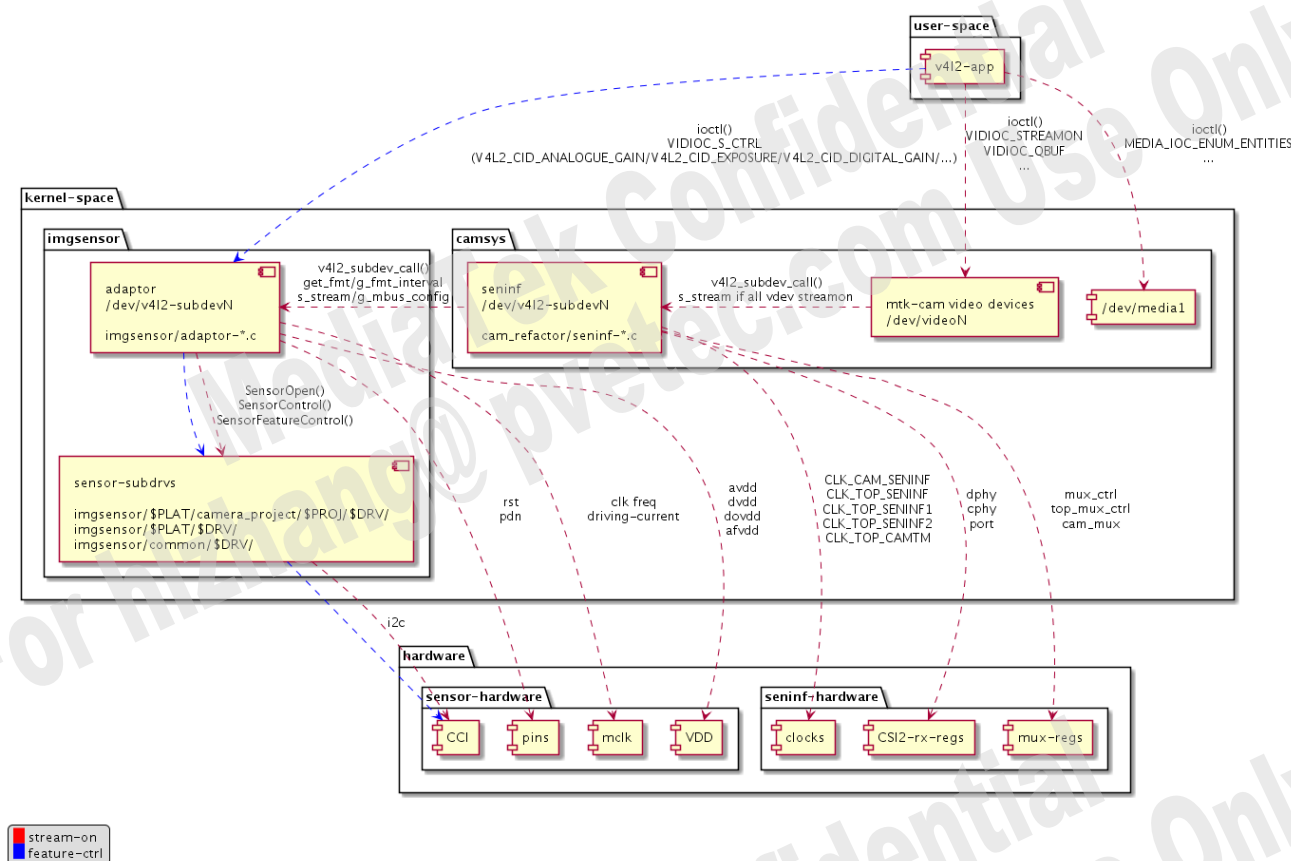
1 Camera Driver

1.1 概述

本章节主要说明 MT8676 Camera 移植方法以及相关问题的调试方法和思路。

1.2 架构/进程概述

Camera driver 主要分 User space 和 Kernel space 两部分，两边通过 IOCTL 传递参数。Raw sensor metadata 等放在 User space；sensor 的相关设定和 power control 是在 Kernel space。



1.3 配置/客制化指南

本节主要介绍 Camera driver 移植指南。

1.3.1 如何添加一个新的 Sensor

这部分对于 YUV 和 Raw sensor 步骤基本一致。

1.3.1.1 移植文件列表

- **Config Files**
 - \device\mediatek\mt6897\CameraConfig.mk
 - \device\mediatek\mt6897\device-camera.mk
 - \kernel\kernel_device_modules-6.1\arch\arm64\configs\mgk_64_k61_defconfig
- **Kernel Space**
 - \kernel\kernel_device_modules-6.1\drivers\misc\mediatek\imgsensor\inc\kd_imgsensor.h
 - \kernel\kernel_device_modules-6.1\arch\arm64\boot\dts\mediatek\cust_mt8676_camera_v4l2.dtsi
 - \vendor\mediatek\kernel_modules\mtkcam\imgsensor\src_v4l2
- **User Space**
 - \device\mediatek\vendor\camera\kernel-headers\kd_imgsensor.h
 - \vendor\mediatek\proprietary\custom\common\hal\imgsensor_metadata\sensor\
 - \vendor\mediatek\proprietary\custom\mt6897\hal\imgsensor_metadata\
 - \vendor\mediatek\proprietary\hardware\mtkcam-core\external\firmware\metadata\

1.3.1.2 修改 Config 文件

- \device\mediatek\mt6897\CameraConfig.mk
 - CUSTOM_HAL_IMGSENSOR = xxxx_mipi_raw xxxx_mipi_raw xxxx_mipi_yuv
 - CUSTOM_KERNEL_IMGSENSOR = xxxx_mipi_raw xxxx_mipi_raw xxxx_mipi_yuv
- \device\mediatek\mt6897\device-camera.mk
 - CUSTOM_HAL_IMGSENSOR = xxxx_mipi_raw xxxx_mipi_raw xxxx_mipi_yuv
- \kernel\kernel_device_modules-6.1\ arch\arm64\configs\mgk_64_k61_defconfig
 - CONFIG_CUSTOM_KERNEL_IMGSENSOR = "xxxx_mipi_raw xxxx_mipi_raw xxxx_mipi_yuv"

在上面三个地方增加新的 sensor name。

1.3.1.3 在 Kernel 中添加 Sensor 文件

- \vendor\mediatek\kernel_modules\mtkcam\imgsensor\src_v4l2\common\\${CamDrv}

参考其他 sensor，在上面目录增加自己的 sensor driver 的目录，命名格式可以参考下图：

> vendor > mediatek > kernel_modules > mtkcam > imgsensord > src-v4l2 > common > max96712_mipi_yuv

Name	Date modified	Type	Size
 Makefile	3/11/2024 12:16 PM	File	1 KB
 max96712mipiyuv_Sensor	3/12/2024 3:05 AM	C Source	25 KB
 max96712mipiyuv_Sensor	3/11/2024 12:16 PM	H File	6 KB

1.3.1.4 添加 Sensor ID 和 Sensor Name

- \device\mediatek\vendor\camera\kernel-headers\kd_imgsensor.h
- \kernel\kernel_device_modules-6.1\drivers\misc\mediatek\imgsensord\inc\kd_imgsensor.h

在以上两个文件内都添加 sensor ID 和 sensor name:

- 添加 sensor ID

```
392 #define MAX96712MIPI_SENSOR_ID 0x52
```

- 添加 sensor name

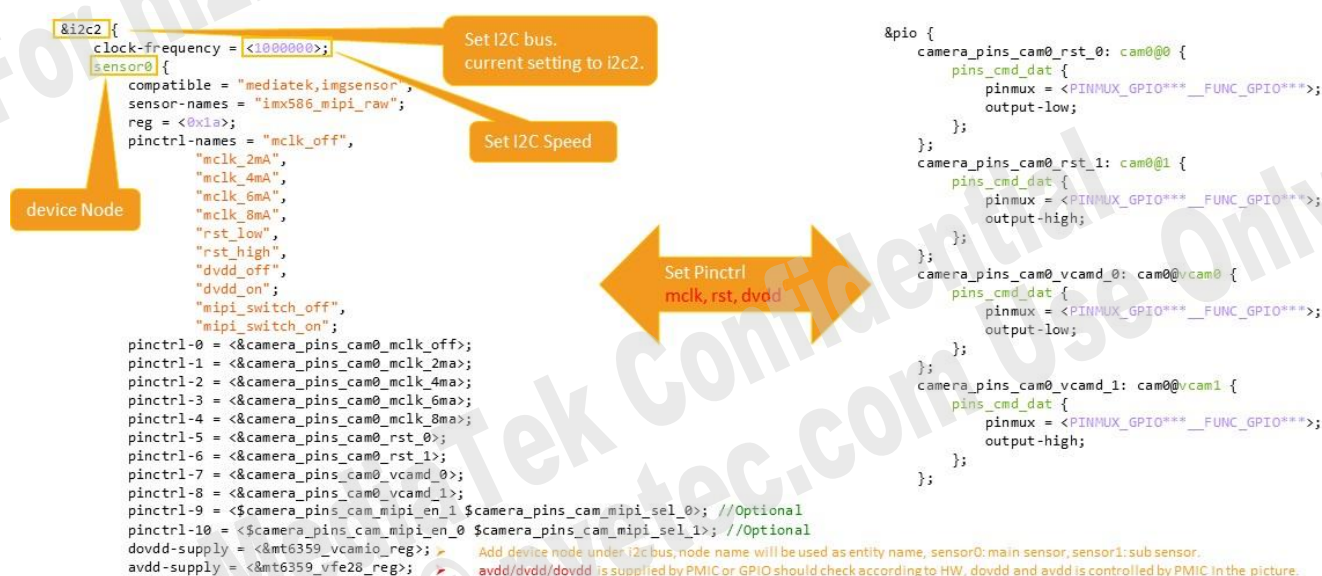
```
642 #define SENSOR_DRVNAME_MAX96712_MIPI_YUV "max96712_mipi_yuv"
```

1.3.1.5 修改 dtb 文件

\kernel\kernel_device_modules-6.1\arch\arm64\boot\dtb\mediatek\cust_mt8676_camera_v4l2.dtsi

通过分析硬件原理图找到当前 sensor 是挂在哪个 I2C 下面, 然后在对应的 I2C 下面增加 sensor 配置, 如下所示:

- 增加了 sensor0

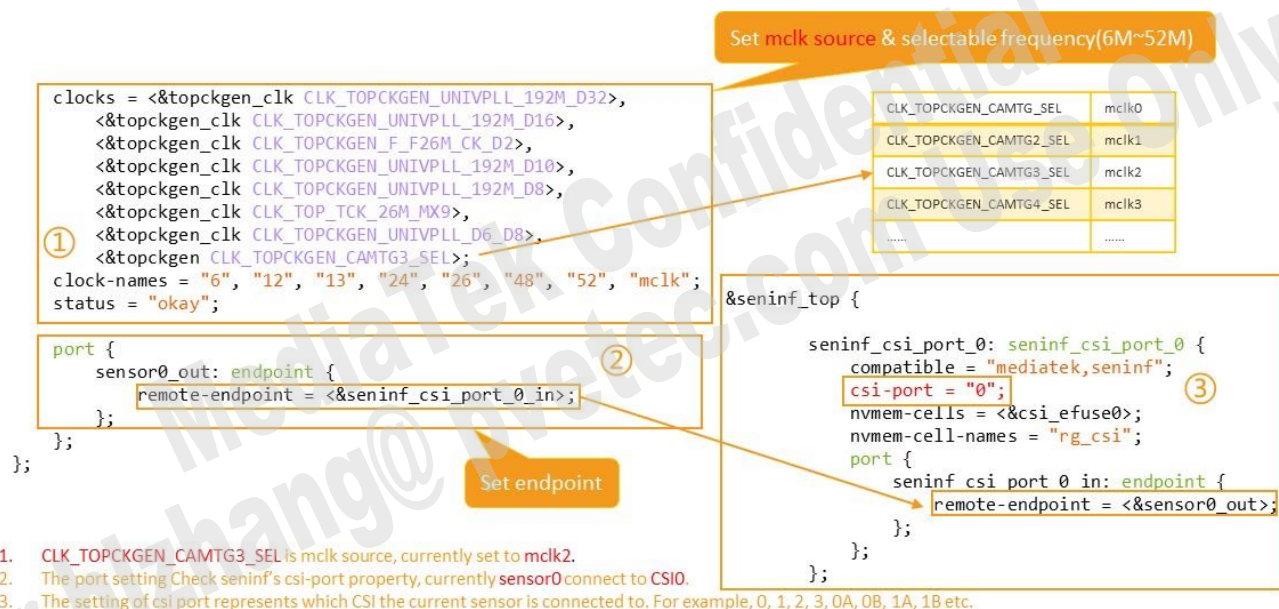


如果更新了 JSON 化功能, 此 sensor 为 raw sensor, 或者是 YUV sensor, 但是不走 JSON 化 flow, 需要修改如下图, 增加 "mediatek, legacy-search" 字段:

```
&i2c7 {
    status = "okay";
    clock-frequency = <1000000>;

    sensor0: sensor0@32 {
        mediatek, legacy-search;
        compatible = "mediatek, imgsensor";
        sensor-names = "ox08b40_mipi_raw";
        reg = <0x52>;
        #thermal-sensor-cells = <0>;
        pdn-supply = <gpio96_regulator>;
    };
};
```

关于 seninf 和 sensor 的关联, 可以参考下图, seninf_top 里 csi-port 的值是硬件实际连接的 csi-port。



与 dts 里配置对应的就是上电时序的部分 (如下所示), 需要参考 sensor 规格书中的上电时序来配置。

1. Add power sequence info

- {power_type, voltage, delay_time in ms}
- Power on sequence from top to bottom
- Power off sequence from bottom to top
- mclk set frequency(24M) and drive current(4mA)

2. Add subdrv entry

- Naming rule: \$(folder-name)_entry
- Macro will generate sensor-list automatically

- Rename a copy according to the existing sensor driver source code, and the part of the register and sensor info should be confirmed with the vendor.

common/imx586_mipi_raw/imx586mipiraw_Sensor.c

```
static struct subdrv_pw_seq_entry pw_seq[] = {
    {HW_ID_MCLK, 24, 0},
    {HW_ID_RST, 0, 0},
    {HW_ID_AVDD, 2800000, 0},
    {HW_ID_AFVDD, 2800000, 0},
    {HW_ID_DVDD, 1100000, 0},
    {HW_ID_DOVDD, 1800000, 1},
    {HW_ID_MCLK_DRIVING_CURRENT, 4, 1},
    {HW_ID_RST, 1, 3},
};

const struct subdrv_entry imx586_mipi_raw_entry = {
    .name = "imx586_mipi_raw",
    .id = IMX586_SENSOR_ID,
    .pw_seq = pw_seq,
    .pw_seq_cnt = ARRAY_SIZE(pw_seq),
    .ops = &ops,
};
```

以 max96712 为例:

```
00838: static struct subdrv_pw_seq_entry pw_seq[] = {
00839:     {HW_ID_PDN, 0, 20},
00840:     {HW_ID_PDN, 1800000, 20},      上电时序
00841: };
00842:
00843: const struct subdrv_entry max96712_mipi_yuv_entry = {
00844:     .name = "max96712_mipi_yuv",
00845:     .id = MAX96712MIPI_SENSOR_ID,
00846:     .pw_seq = pw_seq,
00847:     .pw_seq_cnt = ARRAY_SIZE(pw_seq),  sensor id和sensor name 这里要改
00848:     .ops = &ops,
00849: };
```

1.3.1.6 增加新的 Meta 文件

参考 max96712_mipi_yuv 文件进行修改添加, 可以直接复制, 然后修改文件名和文件内的 sensor name.

- \vendor\mediatek\proprietary\custom\common\hal\imgsensor_metadata\sensor\
- \vendor\mediatek\proprietary\custom\mt6897\hal\imgsensor_metadata\

如需增加新的分辨率, 可以参考以下代码。


```

X:\cas_personal\mtk23977\alps-mp-u0\mp7--2024_06_03_10_00\alps-mp-u0\mp7--2024_06_03_10_00\merged\vendor\mediatek\proprietary\custom\mt6897\hal\imgsensor_metadata\max96712_mipi_yuv\config_static_metadata_project.h
e Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
config_static_metadata_project.h
13 #endif //
14 CONFIG_METADATA_END()
15 //=====
16 CONFIG_METADATA_BEGIN(MTK_SCALER_AVAILABLE_STREAM_CONFIGURATIONS_WITH_DURATIONS)
17 CONFIG_ENTRY_VALUE(NSCam::eImgFmt_BLOB, MINT64)
18 CONFIG_ENTRY_VALUE(1920, MINT64) // width
19 CONFIG_ENTRY_VALUE(1080, MINT64) // height
20 CONFIG_ENTRY_VALUE(MTK_SCALER_AVAILABLE_STREAM_CONFIGURATIONS_OUTPUT, MINT64) // output
21 CONFIG_ENTRY_VALUE(33333333, MINT64) // frame duration
22 CONFIG_ENTRY_VALUE(0, MINT64) // stall duration
23
24 CONFIG_ENTRY_VALUE(NSCam::eImgFmt_RAW16, MINT64)
25 CONFIG_ENTRY_VALUE(1920, MINT64) // width
26 CONFIG_ENTRY_VALUE(1080, MINT64) // height
27 CONFIG_ENTRY_VALUE(MTK_SCALER_AVAILABLE_STREAM_CONFIGURATIONS_OUTPUT, MINT64) // output
28 CONFIG_ENTRY_VALUE(33333333, MINT64) // frame duration
29 CONFIG_ENTRY_VALUE(0, MINT64) // stall duration
30
31 CONFIG_ENTRY_VALUE(NSCam::eImgFmt_YCBCR_420_888, MINT64)
32 CONFIG_ENTRY_VALUE(1920, MINT64) // width
33 CONFIG_ENTRY_VALUE(1080, MINT64) // height
34 CONFIG_ENTRY_VALUE(MTK_SCALER_AVAILABLE_STREAM_CONFIGURATIONS_OUTPUT, MINT64) // output
35 CONFIG_ENTRY_VALUE(33333333, MINT64) // frame duration
36 CONFIG_ENTRY_VALUE(0, MINT64) // stall duration
37
38 CONFIG_ENTRY_VALUE(NSCam::eImgFmt_BLOB, MINT64)
39 CONFIG_ENTRY_VALUE(1920, MINT64) // width
40 CONFIG_ENTRY_VALUE(1080, MINT64) // height
41 CONFIG_ENTRY_VALUE(MTK_SCALER_AVAILABLE_STREAM_CONFIGURATIONS_OUTPUT, MINT64) // output

```

- \vendor\mediatek\proprietary\hardware\mtkcam-core\external\firmware\metadata\

如果更新了 JSON 化功能，YUV sensor 还需要添加 metadata JSON 文件，参考如下，复制一份后修改文件名和文件内的 sensor name 即可。

vendor > mediatek > proprietary > hardware > mtkcam-core > external > firmware > metadata > max96712_mipi_yuv		
Name	Date modified	Type
constructCustRequestMetadata_DEVICE_SENSOR_DRVNAME_MAX96712_MIPI_YUV	3/31/2025 8:06 PM	JSON File
constructCustStaticMetadata_COMMON_SENSOR_SENSOR_DRVNAME_MAX96712_MIPI_YUV	3/31/2025 8:06 PM	JSON File
constructCustStaticMetadata_PLATFORM_AVAILABLE_KEYS_SENSOR_DRVNAME_MAX96712_MIPI_YUV	3/31/2025 8:06 PM	JSON File
constructCustStaticMetadata_PLATFORM_CUSTOM_SENSOR_DRVNAME_MAX96712_MIPI_YUV	3/31/2025 8:06 PM	JSON File
constructCustStaticMetadata_PLATFORM_LENS_SENSOR_DRVNAME_MAX96712_MIPI_YUV	3/31/2025 8:06 PM	JSON File
constructCustStaticMetadata_PLATFORM_MOD_OVERWRITE_SENSOR_DRVNAME_MAX96712_MIPI_YUV	3/31/2025 8:06 PM	JSON File
constructCustStaticMetadata_PLATFORM_MODULE_SENSOR_DRVNAME_MAX96712_MIPI_YUV	3/31/2025 8:06 PM	JSON File
constructCustStaticMetadata_PLATFORM_PRO_OVERWRITE_SENSOR_DRVNAME_MAX96712_MIPI_YUV	3/31/2025 8:06 PM	JSON File
constructCustStaticMetadata_PLATFORM_PROJECT_SENSOR_DRVNAME_MAX96712_MIPI_YUV	3/31/2025 8:06 PM	JSON File
constructCustStaticMetadata_PLATFORM_VENDOR_SENSOR_DRVNAME_MAX96712_MIPI_YUV	3/31/2025 8:06 PM	JSON File
constructCustStaticMetadata_PROJECT_V1_PRO_OVERWRITE_SENSOR_DRVNAME_MAX96712_MIPI_YUV	3/31/2025 8:06 PM	JSON File

1.3.1.7 配置 Sensor Format

根据 bridge 输出的 format，配置 UYVY 或者 YUYV：

```

\vendor\mediatek\kernel_modules\mtkcam\imgsensor\src_v4l2
\common\max96712_mipi_yuv\max96712mipiyuv_Sensor.c

```

查找所有配置 sensor_output_dataformat 的地方改为实际的 format。

```
.sensor_output_dataformat = SENSOR_OUTPUT_FORMAT_UYVY,
```

1.3.1.8 配置 Modestruct

与供应商确认设置的相关参数，并将其填入结构体中。

Pclk->pixel clock, 单位 Hz

Linelength->HTS

Framelength->VTS

$\text{linelength} \times \text{framelength} \times \text{fps} = \text{pclk}$

Grabwindow_width-> camera size 宽 (在 MT8676 上没有这个值, 之后会填在 solution 的位置)

Grabwindow_height->camera size 高 (在 MT8676 上没有这个值, 之后会填在 solution 的位置)

Mipi_pixel_rate-> mipi datarate $\times$ lane_number/bitdepth, YUV bitdepth = 16

```

315 static struct subdrv_mode_struct mode_struct[] = {
316     {
317         .frame_desc = frame_desc_prev,
318         .num_entries = ARRAY_SIZE(frame_desc_prev),
319         .mode_setting_table = PARAM_UNDEFINED,
320         .mode_setting_len = PARAM_UNDEFINED,
321         .seamless_switch_group = PARAM_UNDEFINED,
322         .seamless_switch_mode_setting_table = PARAM_UNDEFINED,
323         .seamless_switch_mode_setting_len = PARAM_UNDEFINED,
324         .hdr_mode = HDR_NONE,
325         .raw_cnt = 1,
326         .exp_cnt = 1,
327         .pclk = 500000000,
328         .linelength = 8096,
329         .framelength = 1400,
330         .max_framerate = 300,
331         .mipi_pixel_rate = 500000000,
332         .readout_length = 0,
333         .read_margin = 10,
334         .framelength_step = 2,
335         .coarse_integ_step = 2,
336         .sensor_output_dataformat = SENSOR_OUTPUT_FORMAT_UYVY,
337         .multi_exposure_shutter_range[IMSENSOR_EXPOSURE_LE].min = 8,
338         .imgsensor_winsize_info = {
339             .full_w = 1920,
340             .full_h = 1280,
341             .x0_offset = 0,
342             .y0_offset = 0,
343             .w0_size = 1920,
344             .h0_size = 1280,
345             .scale_w = 1920,
346             .scale_h = 1280,
347             .x1_offset = 0,
348             .y1_offset = 0,
349             .w1_size = 1920,
350             .h1_size = 1280,
351             .x2_tg_offset = 0,
352             .y2_tg_offset = 0,
353             .w2_tg_size = 1920,
354             .h2_tg_size = 1280,
355         },
356         .pdaf_cap = TRUE,
357         .imgsensor_pd_info = NULL,
358         .ae_binning_ratio = 1465,
359         .fine_integ_line = 826,
360         .delay_frame = 3,
361         .csi_param = {
362             .legacy_phy = 0,
363             .not_fixed_trail_settle = 1,
364             .not_fixed_dphy_settle = 1,
365             .dphy_trail = 40,
366             .dphy_data_settle = 8,
367             .cphy_settle = 13,
368             .clk_lane_no_initial_flow = 0,
369         },
370         .dpc_enabled = true, /* reg 0x0b06 */
371     }, /* preview */
372     {
373         .frame_desc = frame_desc_prev,

```

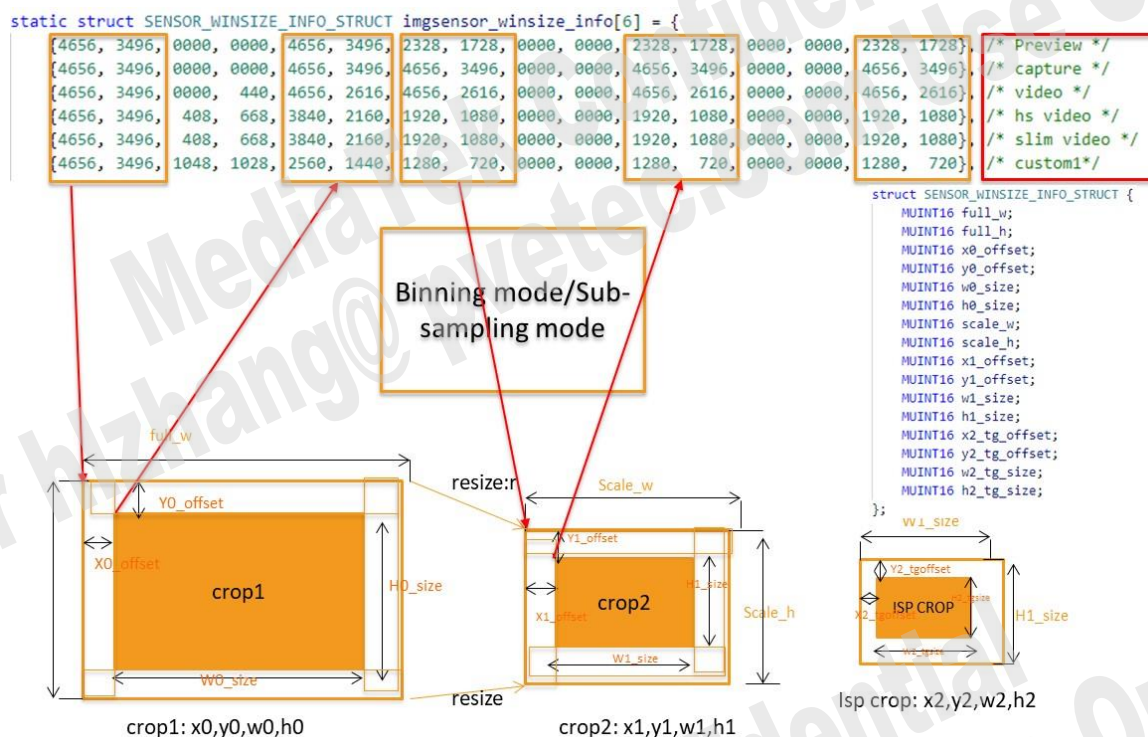
pclk, linelength, framelength
framerate, mipi pixel rate 在这里配置

format 在这里配置

winsize info 在这里配置

hs trail & settle 在这里配置

下图是借用了之前版本的一个对 winsize info 的解释，来描述各个字段的含义



限制: full size (4:3) 或 full size (16:9) 的宽和高需要进行 4 倍数对齐。

配置 sensor 输出的宽高，与 Grabwindow width/Grabwindow height 的大小一致。

Grab window 要求:

1. 设置的宽必须是 **16** 的倍数，高必须为 **4** 的倍数。
2. 相同比例下视角一致：4:3 与 16:9 要求 sensor 输出的 window 保证水平方向和 full size 视角一致。
3. Grab window 设置建议和 sensor 输出的 window 一致。

特殊情况下是: $\text{grabwindow width}(\text{height}) \leq \text{sensor output width}(\text{height}) - \text{startx}(\text{y})$

1.3.1.9 配置 static ctx

如下图所示:

1. `sensor_id` 字段需填对应的 sensor ID。
2. `i2c_add_table` 一般只配置 0x52 即可，即实际的解串器的地址。
3. 配置分辨率，这里就是前面提到的 `grabwindow` 的值
4. 配置 `mipi_lane_num`，这里表示是 MIPI 的 lane 的个数。
5. `cam_type` 是按照模组场景的类型填写，如果是单 sensor 的话，可以填 `MTK_SENSOR_TYPE_SINGLE`，它有如下值可以选择以下几种：
 - `SENSOR_TYPE_COMB_AVM` 表示组帧 `avm`

- SENSOR_TYPE_SINGLE 表示单路 VC sensor
- SENSOR_TYPE_MULTI 表示 multicam
- SENSOR_TYPE_MULTI_ASYNC 表示各路 VC 相互独立不影响的 multicam
- SENSOR_TYPE_NON_COMB_AVM 表示非组帧 avm

如配置为 SENSOR_TYPE_MULTI, 还需配置 group_info 如下:

```
.group_info = {
    4, master sensorId
    4, vc num
    {
        {6,0,0}, slave sensorId, vc_id, link status
        {7,1,0},
        {8,2,0},
        {9,3,0},
    },
},
```

如果配置为 SENSOR_TYPE_MULTI_ASYNC, 则还需要配置 group_dt_info 如下:

推荐使用 SENSOR_TYPE_MULTI_ASYNC, SLAVE 驱动之间不会相互影响。

```
.group_dt_info = {
    4, // VC NUM
    {
        {0,0x1E}, // VC_ID & dt
        {1,0x1E},
        {2,0x1E},
        {3,0x1E},
    },
},
```

6. sensor_output_dataformat 表示模组输出的数据的 format, 填实际的 format 即可

```
00571: static struct subdrv_static_ctx static_ctx = {
00572:     .sensor_id = MAX96712MIPI_SENSOR_ID,
00573:     .reg_addr_sensor_id = {0x0000},
00574:     .i2c_addr_table = {0x52, 0xFF},
00575:     .i2c_burst_write_support = TRUE,
00576:     .i2c_transfer_data_type = I2C_DT_ADDR_16_DATA_8,
00577:     .eeprom_info = 0,
00578:     .eeprom_num = 0,
00579:     .resolution = {1920, 1280},
00580:     .mirror = IMAGE_HV_MIRROR,
00581:
00582:     .mclk = 24,
00583:     .isp_driving_current = ISP_DRIVING_4MA,
00584:     .sensor_interface_type = SENSOR_INTERFACE_TYPE_MIPI,
00585:     .mipi_sensor_type = MIPI_OPHY_NCSI2,
00586:     .mipi_lane_num = SENSOR_MIPI_4_LANE,
00587:     .ob_pedestal = 0x40,
00588:     .cam_type = SENSOR_TYPE_NON_COMB_AVM,
00589:     .sensor_output_dataformat = SENSOR_OUTPUT_FORMAT_UYVY,
```

配置分辨, 这里就是前面grabWindow的值

1.3.1.10 配置 VC 信息

如果是单个 VC，可以进行如下配置：

```
00303:
00304: static struct mtk_mbus_frame_desc_entry frame_desc_prev[] = {
00305:     {
00306:         .bus.csi2 = {
00307:             .channel = 0,
00308:             .data_type = 0x1E, //yuv422
00309:             .hsize = FRAME_WIDTH,
00310:             .vsize = FRAME_HEIGHT,
00311:             .user_data_desc = VC_BRIDGE_RAW_0,
00312:         },
00313:     },
00314: };
```

如果是有多多个 VC，可以进行如下配置，主要是修改 channel 和 user_data_desc：

```
00304: static struct mtk_mbus_frame_desc_entry frame_desc_prev[] = {
00305:     {
00306:         .bus.csi2 = {
00307:             .channel = 0,
00308:             .data_type = 0x1E, //yuv422
00309:             .hsize = FRAME_WIDTH,
00310:             .vsize = FRAME_HEIGHT,
00311:             .user_data_desc = VC_BRIDGE_RAW_0,
00312:         },
00313:     },
00314:     {
00315:         .bus.csi2 = {
00316:             .channel = 1,
00317:             .data_type = 0x1E, //yuv422
00318:             .hsize = FRAME_WIDTH,
00319:             .vsize = FRAME_HEIGHT,
00320:             .user_data_desc = VC_BRIDGE_RAW_1,
00321:         },
00322:     },
00323:     {
00324:         .bus.csi2 = {
00325:             .channel = 2,
00326:             .data_type = 0x1E, //yuv422
00327:             .hsize = FRAME_WIDTH,
00328:             .vsize = FRAME_HEIGHT,
00329:             .user_data_desc = VC_BRIDGE_RAW_2,
00330:         },
00331:     },
00332:     {
00333:         .bus.csi2 = {
00334:             .channel = 3,
00335:             .data_type = 0x1E, //yuv422
00336:             .hsize = FRAME_WIDTH,
00337:             .vsize = FRAME_HEIGHT,
00338:             .user_data_desc = VC_BRIDGE_RAW_3,
00339:         },
00340:     },
00341: };
```

1.3.1.11 配置获取 Sensor ID 函数

```
00249: static int return_sensor_id(struct subdrv_ctx *ctx)
00250: {
00251:     return read_cmos_sensor(ctx, MAX96712_ID_REG);
00252: }
00253:
```

1.3.1.12 配置 Sensor Init 函数

下面的函数内主要是配置模组的设置，需要客户和供应商调试生成。

```
88
89 static int max96712_sensor_init(struct subdrv_ctx *ctx)
90 {
91
92     DRV_LOG(ctx, "--MAX96712-ISX019-sensor_init start\n");
93     write_cmos_sensor(ctx, 0x0017, 0x14);
94     write_cmos_sensor(ctx, 0x0019, 0x94);
95     write_cmos_sensor(ctx, 0x0013, 0x75);
```

1.3.1.13 配置 get_sensor_usage 函数

```
668
669 static int get_sensor_usage(struct subdrv_ctx *ctx, enum mtk_sensor_usage *usage)
670 {
671     *usage = MTK_SENSOR_USAGE_NONCOMB;
672     return 0;
673 }
```

它的值可以选择下面几种，COMB 表示组帧 AVM，MUTI 表示 multicam，MUTI_ASYNC 表示各路 VC 独立不相互影响的 multicam，NONCOMB 表示非组帧 AVM，单 sensor 的话配置为 SINGLE。

```
42 enum mtk_sensor_usage {
43     MTK_SENSOR_USAGE_SINGLE,
44     MTK_SENSOR_USAGE_COMB,
45     MTK_SENSOR_USAGE_MUTI,
46     MTK_SENSOR_USAGE_NONCOMB,
47     MTK_SENSOR_USAGE_MUTI_ASYNC,
48 };
```

1.3.1.14 配置 Streaming Control

初始化的设定全部放到 sensor_init() 函数里。MIPI enable 和 disable 函数放到 streaming_control() 函数中。

```

221
222 static int set_streaming_control(void *arg, bool enable)
223 {
224     struct subdrv_ctx *ctx = (struct subdrv_ctx *)arg;
225
226     DRV_LOG(ctx, "streaming_enable(0=Sw Standby,1=streaming): %d\n", enable);
227     if (enable) {
228         /*turn on mipi*/
229         write_cmos_sensor(ctx, 0x040B, 0x42);
230         mdelay(5);
231     } else {
232         /*turn off mipi*/
233         write_cmos_sensor(ctx, 0x040B, 0x0);
234         mdelay(5);
235     }
236
237     return ERROR_NONE;
238 }

```

1.3.1.15 subdrv_static_ctx 的成员解析

Member	Meaning
sensor_id	Sensor ID defined in kd_imgsensor.h.
reg_addr_sensor_id	Sensor register address where sensor ID is read. Up to 3 bytes.
i2c_addr_table	I2C write ID, end in 0xff, 4 for maximum. E.g., i2c_addr_table = {0x20, 0x6e, 0xff}, .
eeeprom_info	The address of eeeprom_info_struct. If there is no eeeprom device, set to 0.
eeeprom_num	Array size of the eeeprom_info_struct. If there is no eeeprom device, set to 0.
Resolution	Full pixel size of sensor output.
mirror	Set IMAGE_HV_MIRROR if the sensor orientation is 180 degrees difference from module orientation.
mclk	Overwrite MCLK frequency (MHz)
isp_driving_current	Overwrite MCLK driving current (mA)
sensor_interface_type	Data output interface type. Set default MIPI.
mipi_sensor_type	C-PHY or D-PHY.
mipi_lane_num	How many lanes/trios of MIPI-PHY.
Ob_pedestal	OB offset. General is 64.
sensor_output_dataformat	Bayer order, 4-cell order and SW/HW remo.
ana_gain_def	Default analog gain. Set 4x. 1024base.
ana_gain_min	Minimum analog gain. 1024base.
ana_gain_max	Maximum analog gain. 1024base.
ana_gain_type	Sony:type 0; OV:type 1; Samsung:type 2; Hinyx:type 3; GC:type 4
ana_gain_step	Minimum valid step of analog gain. 1024base.
ana_gain_table	Use valid analog gain table. Remove unsuitable value which is not linearity. 1024base.
ana_gain_table_size	Size of analog gain table.
min_gain_iso	Set minimum ISO 100.
exposure_def	Default exposure line. Set 0x3D0.

Member	Meaning
exposure_min	Minimum exposure line.
exposure_max	Maximum exposure line.
exposure_step	Minimum valid step of exposure line.
exposure_margin	Maximum margin of exposure line.
frame_length_max	Maximum framelength
ae_effective_frame	AE effective frame
frame_time_delay_frame	The frame “frame length” setting takes effect. Sony sensor filled in 3; other sensors filled in 2.
start_exposure_offset	Parameter tuned in CTS sensor fusion test.
pdaif_type	Reference enum IMGSENSOR_PDAF_SUPPORT_TYPE_ENUM.
hdr_type	Reference enum IMGSENSOR_HDR_SUPPORT_TYPE_ENUM.
seamless_switch_support	If sensor supports seamless switch, set 1 to enable seamless switch function.
temperature_support	Set to 1 if sensor supports temperature sensor readout.
g_temp	Implement get temperature function.
g_gain2reg	Implement analog gain to register mapping rule.
s_gph	Implement set enable/disable group hold function.
s_cali	Implement the function to write data from eeprom_info_struct to sensor (e.g., QSC, cross-talk, ...)
reg_addr_stream	Sensor register address where streaming on/off is controlled.
Reg_addr_mirror_flip	Sensor register address where mirror/flip on/off is controlled. No use when set to 0.
reg_addr_exposure	Sensor register address where exposure line is set. Up to 3 channels of exposure. Up to 3 bytes.
long_exposure_support	Set to 1 if sensor supports long exposure left shift function.
reg_addr_exposure_lshift	Sensor register address where long exposure left shift is set.
reg_addr_ana_gain	Sensor register address where analog gain is set. Up to 3 channels of exposure. Up to 3 bytes.
reg_addr_frame_length	Sensor register address where framelength is set.
reg_addr_temp_en	If “temperature_support” is set to 1. Sensor register address where temperature sensor on/off is controlled.
reg_addr_temp_read	If “temperature_support” is set to 1. Sensor register address where temperature output is read.
reg_addr_auto_extend	For Sony sensor, sensor register address where auto extend function enable/disable is controlled.
reg_addr_frame_count	For Samsung sensor, sensor register address where frame count is read to determine whether sensor stream is off or not.
reg_addr_fast_mode	For Sony sensor, sensor register address where fast mode on/off is controlled.
init_setting_table	The address of sensor initial setting table
init_setting_len	Array size of the sensor initial setting table

Member	Meaning
mode	The address of subdrv_mode_struct
sensor_mode_num	Array size of the subdrv_mode_struct
list	The address of customized feature control list
list_len	Array size of the customized feature control list
checksum_value	The calculated value when Test Pattern output, for Camera Auto Test

1.3.1.16 配置 Sensor Mode

Subdrv_mode_struct mode_struct[] 这个结构体数组是用来存不同 Sensor Mode 的数据配置的。

下面以 Preview 为例子说明了一些需要客户修改的字段，对于 YUV sensor，一般后面的 Sensor Mode 和 Preview 是完全一样的。对于 Raw sensor 可能会有差异，根据实际需要来配置。

```

315 static struct subdrv_mode_struct mode_struct[] = {
316 *
317 *     .frame_desc = frame_desc_prev,
318 *     .num_entries = ARRAY_SIZE(frame_desc_prev),
319 *     .mode_setting_table = PARAM_UNDEFINED,
320 *     .mode_setting_len = PARAM_UNDEFINED,
321 *     .seamless_switch_group = PARAM_UNDEFINED,
322 *     .seamless_switch_mode_setting_table = PARAM_UNDEFINED,
323 *     .seamless_switch_mode_setting_len = PARAM_UNDEFINED,
324 *     .hdr_mode = HDR_NONE,
325 *     .raw_cnt = 1,
326 *     .exp_cnt = 1,
327 *     .pclk = 500000000,
328 *     .linelength = 8096,
329 *     .framelength = 1400,
330 *     .max_framerate = 300,
331 *     .mipi_pixel_rate = 500000000,
332 *     .readout_length = 0,
333 *     .read_margin = 10,
334 *     .framelength_step = 2,
335 *     .coarse_integ_step = 2,
336 *     .sensor_output_dataformat = SENSOR_OUTPUT_FORMAT_UYVY,
337 *     .multi_exposure_shutter_range[IMGSSENSOR_EXPOSURE_LE].min = 0,
338 *     .imgsensor_winsize_info = {
339 *         .full_w = 1920,
340 *         .full_h = 1280,
341 *         .x0_offset = 0,
342 *         .y0_offset = 0,
343 *         .w0_size = 1920,
344 *         .h0_size = 1280,
345 *         .scale_w = 1920,
346 *         .scale_h = 1280,
347 *         .x1_offset = 0,
348 *         .y1_offset = 0,
349 *         .w1_size = 1920,
350 *         .h1_size = 1280,
351 *         .x2_tg_offset = 0,
352 *         .y2_tg_offset = 0,
353 *         .w2_tg_size = 1920,
354 *         .h2_tg_size = 1280,
355 *     },
356 *     .pdaf_cap = TRUE,
357 *     .imgsensor_pd_info = NULL,
358 *     .ae_binning_ratio = 1465,
359 *     .fine_integ_line = 826,
360 *     .delay_frame = 3,
361 *     .csi_param = {
362 *         .legacy_phy = 0,
363 *         .not_fixed_trail_settle = 1,
364 *         .not_fixed_dphy_settle = 1,
365 *         .dphy_trail = 40,
366 *         .dphy_data_settle = 8,
367 *         .dphy_settle = 13,
368 *         .clk_lane_no_initial_flow = 0,
369 *     },
370 *     .dpc_enabled = true, /* reg 0x0b06 */
371 * }, /* preview */
372 *

```

pclk, linelenth, framelength
framerate, mipi pixel rate 在这里配置

format 在这里配置

winsize info 在这里配置

hs trail & settle 在这里配置

1.3.1.17 Driver 功能

1.3.1.17.1 Driver 操作函数 list

Driver 函数的 list 如下:

```
789 static struct subdrv_ops ops = {
790     .get_id = get_imgsensor_id,
791     .init_ctx = init_ctx,
792     .open = open,
793     .get_info = common_get_info,
794     .get_resolution = common_get_resolution,
795     .control = common_control,
796     .feature_control = common_feature_control,
797     .close = common_close,
798     .get_frame_desc = common_get_frame_desc,
799     .get_temp = common_get_temp,
800     .get_csi_param = common_get_csi_param,
801     .vsync_notify = vsync_notify,
802     .update_sof_cnt = common_update_sof_cnt,
803     .parse_ebd_line = common_parse_ebd_line,
804     .get_sensor_usage = get_sensor_usage,
805 };
806
```

在以 adapter 名字开头的 imgsensor 的代码里, 其往往是通过 subdrv_call 的方式来调用到 driver 代码里的, 就是上面的函数 list, 比如:

```
ret = subdrv_call(ctx, get_id, &sensor_id);
```

上面显示的调用就会 call 到上面函数 list 里的 get_sensor_id。

上面 list 里有很多 common 开头的函数, 这个部分是用的通用的处理流程。不需要在 sensor driver 里特别实现, 下面就不做过介绍了。

针对贵司可能改到的 get_imgsensor_id 和 open 函数, 请参考下面两个章节。

1.3.1.17.2 get_imgsensor_id 函数

开机 search sensor 时会通过 get_imgsensor_id 函数读取 ID。若能成功读到 ID, 在 UI 上显示 Camera APP 的图标。

下面函数主要有两个功能: 一获取 sensor ID, 二判断是否有连接到模组

```

675 static int get_imgsensor_id(struct subdrv_ctx *ctx, u32 *sensor_id)
676 {
677     u8 i = 0;
678     u8 retry;
679
680     ctx->i2c_write_id = static_ctx.i2c_addr_table[0];
681     retry = 2;
682
683     while (static_ctx.i2c_addr_table[i] != 0xff) {
684         spin_lock(&imgsensor_drv_lock);
685         ctx->i2c_write_id = static_ctx.i2c_addr_table[i];
686         spin_unlock(&imgsensor_drv_lock);
687         do {
688             *sensor_id = return_sensor_id(ctx);
689             if (*sensor_id == (static_ctx.sensor_id)) {
690                 DRV_LOG(ctx, "i2c write id: 0x%x, sensor id: 0x%x\n",
691                     ctx->i2c_write_id, *sensor_id);
692                 return ERROR_NONE;
693             }
694             DRV_LOG(ctx, "Read id fail, id: 0x%x, i2c id: 0x%x\n",
695                 *sensor_id, ctx->i2c_write_id);
696             retry--;
697         } while (retry > 0);
698         i++;
699         retry = 2;
700     }
701     if (*sensor_id != (static_ctx.sensor_id)) {
702         *sensor_id = 0xFFFFFFFF;
703         return ERROR_SENSOR_CONNECT_FAIL;
704     }
705     return ERROR_NONE;
706 }
707

```

1.3.1.17.3 Open 函数

每次进 camera 时会调用。

读 sensor_id，确认 I2C 通信是否正常。

调用 sensor_init 函数初始化 ctx 结构体中的一些变量。

```

688 static int open(struct subdrv_ctx *ctx)
689 {
690     u8 i = 0;
691     u8 retry = 2;
692     u16 sensor_id = 0;
693     u32 scenario_id = 0;
694
695     while (static_ctx.i2c_addr_table[i] != 0xff) {
696         spin_lock(&imgsensor_drv_lock);
697         ctx->i2c_write_id = static_ctx.i2c_addr_table[i];
698         spin_unlock(&imgsensor_drv_lock);
699         do {
700             sensor_id = return_sensor_id(ctx);
701             if (sensor_id == MAX96712ISX_SENSOR_ID) {
702                 DRV_LOG(ctx, "i2c write id: 0x%x, sensor id: 0x%x\n",
703                     ctx->i2c_write_id, sensor_id);
704                 break;
705             }
706             DRV_LOG(ctx, "Read id fail, id: 0x%x, i2c id: 0x%x\n",
707                 sensor_id, ctx->i2c_write_id);
708             retry--;
709         } while (retry > 0);
710         i++;
711         if (sensor_id == MAX96712ISX_SENSOR_ID)
712             break;
713         retry = 2;
714     }
715     if (sensor_id != MAX96712ISX_SENSOR_ID)
716         return -ERROR_SENSOR_CONNECT_FAIL;
717
718     /* initail sequence write in */
719     max96712_sensor_init(ctx);
720     retry = 3;
721     do {
722         if (Chenck_Sensor_Signal(ctx)) {
723             DRV_LOG(ctx, "Sensor Not Connect");
724             mdelay(5);
725             retry--;
726         } else
727             break;
728     } while (retry > 0);
729
730     if (retry == 0) {
731         DRV_LOG(ctx, "OPEN FAIL!Sensor Not Connect");
732         return -ERROR_SENSOR_CONNECT_FAIL;
733     }
734 }
735
736 /* move camera init to initctx(), void max9286 init too much time */
737
738 spin_lock(&imgsensor_drv_lock);
739
740 memset(ctx->exposure, 0, sizeof(ctx->exposure));
741 memset(ctx->ana_gain, 0, sizeof(ctx->gain));
742 ctx->exposure[0] = ctx->s_ctx.exposure_def;
743 ctx->ana_gain[0] = ctx->s_ctx.ana_gain_def;
744 ctx->current_scenario_id = scenario_id;
745 ctx->pclk = mode_struct[scenario_id].pclk;
746 ctx->line_length = mode_struct[scenario_id].linelength;
747 ctx->frame_length = mode_struct[scenario_id].framelength;
748 ctx->frame_length_rg = ctx->frame_length;
749 ctx->current_fps = ctx->pclk / ctx->line_length * 10 / ctx->frame_length;
750 ctx->readout_length = mode_struct[scenario_id].readout_length;
751 ctx->read_margin = mode_struct[scenario_id].read_margin;
752 ctx->min_frame_length = ctx->frame_length;
753 ctx->autoflicker_en = FALSE;
754 ctx->test_pattern = 0;
755 ctx->ihdr_mode = 0;
756 ctx->pdaf_mode = 0;
757 ctx->hdr_mode = 0;
758 ctx->extend_frame_length_en = 0;
759 ctx->is_seamless = 0;
760 ctx->fast_mode_on = 0;
761 ctx->sof_cnt = 0;
762 ctx->ref_sof_cnt = 0;
763 ctx->is_streaming = 0;
764 spin_unlock(&imgsensor_drv_lock);
765
766 return ERROR_NONE;
767 }

```

批注:

1. 关于分辨率的说明

由于场景不同，给到 User space 的图的大小和实际 seninf 收取的 VC 的大小不同，所以在 Driver 里会有两个分辨率，客户可以以 max96712isx 这个非组帧 avm 的 Driver 为例。

搜索下面的宏进行对照配置。

```

00025:
00026: #define VC 0
00027:
00028: #define SUPERFRAME_WIDTH 7680
00029: #define SUPERFRAME_HEIGHT 1280
00030:
00031: #define FRAME_WIDTH 1920
00032: #define FRAME_HEIGHT 1280
00033:
00034: #define MAX96712_ID_REG 0x000
00035: #define SC22AT_I2C_ADDR 0x60
00036: #define MAX96717_I2C_ADDR 0x80
00037: #define MAX96712_SENSOR_ID 0x52
00038:
00039: #define LOG_INF(format, args...) pr_info(PFX "[%s] " format, __func__, ##args)
00040: #define LOG_DBG(format, args...) pr_debug(PFX "[%s] " format, __func__, ##args)
00041: #define LOG_ERR(format, args...) pr_info(PFX "[%s] " format, __func__, ##args)
00042: #define read_cmos_sensor(...) subdrv_i2c_rd_u8(__VA_ARGS__)
00043: #define write_cmos_sensor(...) subdrv_i2c_wr_u8(__VA_ARGS__)
00044:
00045: #define read_cmos_sensor_with_i2cid(subctx, i2c_id, reg) \
00046: ({

```

这是上层看到的最大分辨率，如非组帧AVM，这里的width会是7680

这是单个VC的分辨率

2. 关于 dts 里 GPIO 供电的部分，还可以选择 regulator 封装的方式。

用法如下：

```

gpio141_regulator: gpio-141 {
    compatible = "regulator-fixed";
    regulator-name = "gpio141_regulator";
    regulator-min-microvolt = <1800000>;
    regulator-max-microvolt = <1800000>;
    gpio = <&pio 141 0x0>;
    enable-active-high;
    status = "okay";
};

sensor0: sensor0@52 {
    compatible = "mediatek,imgsens0r";
    sensor-names = "max96712_mipi_yuv";
    reg = <0x52>;
    #thermal-sensor-cells = <0>;
    pdn-supply = <&gpio141_regulator>;

    clocks = <&topckgen_clk CLK_TOP_UNIVPLL_192M_D32>,
        <&topckgen_clk CLK_TOP_UNIVPLL_192M_D16>,
        <&topckgen_clk CLK_TOP_F26M_CK_D2>,
        <&topckgen_clk CLK_TOP_UNIVPLL_192M_D10>,
        <&topckgen_clk CLK_TOP_UNIVPLL_192M_D8>,
        <&topckgen_clk CLK_TOP_UNIVPLL_D6_D16>,
        <&topckgen_clk CLK_TOP_UNIVPLL_D6_D8>,
        <&topckgen_clk CLK_TOP_CAMTG5_SEL>;
    clock-names = "6", "12", "13", "19.2", "24", "26", "52", "mclk";

    status = "okay";

    port {
        sensor0_out: endpoint {
            remote-endpoint = <&seninf_csi_port_2_in>;
        };
    };
};

```

3. UT 测试指令

```
C:\Users>adb shell sentest_v4l2 //just search sensor, not streaming
```

```
[show_Sensors]sensorNum 2
[show_Sensors]name:SENSOR_DRVNAME_OV05A20_MIPI_RAW type:0
[show_Sensors]index:0, SensorDevIdx:1
[show_Sensors]name:SENSOR_DRVNAME_IMX576_MIPI_RAW type:0
[show_Sensors]index:1, SensorDevIdx:2
[main]Param: 1 <sensorDev> <scenario> <fps>
[main]<sensorDev> : main(1), Sub(2), Main2(4), sub2(8), Main3(16)
[main]<scenario> : Pre(0), Cap(1), VD(2), slim1(3), slim2(4)
```

显示 sensorNum 不为 0，并且展示了传感器的具体名称，说明传感器搜索成功，接下来测试传感器的输出。

```
C:\Users> adb shell sentest_v4l2 1 0 //search sensor, then open main (1) sensor preview (0)
```

```
[test_SensorInterface]name:SENSOR_DRVNAME_MAX96712ISX_MIPI_YUV type:1
[test_SensorInterface]index:0, SensorDevIdx:1
[test_SensorInterface]scenario(0) , binning type = 1465
[test_SensorInterface]scenario(0) , rawFmtType = 0
[test_SensorInterface]scenario(0) , OB_pedestal = 64
[test_SensorInterface]scenario(0) , sensorFormatOrder = 4
[test_SensorInterface]scenario(0) , seamless_targets_count = 0
[test_SensorInterface]Sensor index:0, scenarioId:0
[test_SensorInterface]crop w:1280, h:720
[test_SensorInterface]HDR_Support:0
[test_SensorInterface]scenario(0) , RGBW_Support = 0
[test_SensorInterface]scenario(0) , PDAF_Support = 0, dpc(1), pdc(0)
[test_SensorInterface]scenario(0) , Def_MirrorFlip = 3
[test_SensorInterface]scenario(0) , sensorFormatCellType = 0
[test_SensorInterface]SENSOR_CMD_GET_SENSOR_CROP_WIN_INFO ret 0
[test_SensorInterface]SENSOR_CMD_GET_SENSOR_RGBW_OUTPUT_MODE rgbw mode = 0
[test_SensorInterface]ana gain min/max = 1024/65536
[test_SensorInterface]dig gain min/max = 1024/16384
[test_SensorInterface]dig gain step = 4
[test_SensorInterface]SENSOR_CMD_GET_OFFSET_TO_START_OF_EXPOSURE exposure_offset = 500000
[test_SensorInterface]pHalSensor->powerOn_start
[test_CID][test g_volatile_ctrl] scenario (by set format)
[test_CID][V4L2_CID_VBLANK] ret:0 VB:680
[test_CID][V4L2_CID_HBLANK] ret:0 HB:4434
[test_CID][V4L2_CID_MTK_SENSOR_PIXEL_RATE] ret:0 pixel rate:200000000
[test_CID][V4L2_CID_MTK_STAGGER_INFO] ret:0 count:0 order:
[test_CID][V4L2_CID_MAX_EXP_TIME]ret 0 exposure 0, max_exp_time 0
[test_CID][V4L2_CID_MAX_EXP_TIME]ret 0 exposure 1, max_exp_time 0
[test_CID][V4L2_CID_MAX_EXP_TIME]ret 0 exposure 2, max_exp_time 0
[streamon]sensor: idx=0 devpath=/dev/v4l-subdev26 drvname=max96712isx_mipi_yuv
[streamon]seninf: devpath=/dev/v4l-subdev4 cid=0x98f901 en=1
[test_SensorInterface]main/rolling_shutter=20571428
[test_SensorInterface]main/exp_margin=1371
[test_SensorInterface]main/atv-Continous=1
[test_SensorInterface]main/atv-BypassSensorScenario=0
[test_SensorInterface]main/min_shutter=24 shutter_step=2
[test_SensorInterface]TgInfo[0] = 0
[test_SensorInterface]pixelMode[0] = 0
[test_SensorInterface]mipi rate = 200000000
[test_SensorInterface]wait enter key to end sensorIndex:0...
```

查看 seninf 状态的指令如下，查到结果后，可以发给联发科技确认。

```
cat /sys/devices/platform/soc/1a00e000.seninf-top/status
```



```

255|auto8676p1_64_bsp:/ # cat /sys/devices/platform/soc/1a00e000.seninf-top/status

clk-fmeter-isp(687576) clk-fmeter-csi0(26000) clk-fmeter-csi1(0) clk-fmeter-csi2(0) clk-fmeter-csi3(312000) clk-fmeter-csi4(0) clk-fmeter-csi5(0)

[seninf-8] port 3 intf 6 test 0 cphy 0 lanes 0

[seninf-7] port 3 intf 6 test 0 cphy 0 lanes 0
source sensor7 flags 0x4

[seninf-6] port 3 intf 6 test 0 cphy 0 lanes 4
source sensor6 flags 0x5
csirx_mac_csi irq_stat 0x003f0324, seninf_irq_stat 0x00000000
csi2_line_frame_num 0x00000000
csi2_packet_status 0x12c0012b
csi2_packet_cnt_status 0x000004fb
rx-ana_settle_ck 0x00 dt 0x1a
rx-ana_trail_en 0 param 0x00
CSIRX_MAC_CSI2_RESYNC_MERGE_CTRL 0x00000106 CDPHY_RX_ANA_SETTING_1 0x04340244 DPHY_RX_SPARE0 0x000000f1
data_not_enough_cnt : <0>
err_lane_resync_cnt : <0>
crc_err_cnt : <0>
ecc_err_double_cnt : <0>
ecc_err_corrected_cnt : <0>
after_write_clear_csi_irq 0x0 seninf_irq_stat 0x00000000
[0] vc 0x0 dt 0x2b mux 1 cam 0
mux[1] en 1 src 6 irq_stat 0xc
fifo_overrun_cnt : <0>
cam[0] en 1 Src 8 exp 0x8700f00 res 0x8700f00 err 0x0 irq_stat 0x100
size_err_cnt : <0>
--- mux meter ---
width 3840 height 2160
h_valid 0, h_blank 0
v_valid 0, v_blank 0
mipi_pixel_rate 0
v_blank 0 us
h_blank 0 us
line_time 0 us

[seninf-2] port 2 intf 4 test 0 cphy 0 lanes 0

[seninf-1] port 1 intf 2 test 0 cphy 0 lanes 0
auto8676p1_64_bsp:/ #

```

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